

Deep Learning, Recognition, Feature Detection, Motion Estimation and Computational Photography

Detailed Notes (Exam-Oriented)

1. Deep Learning

Definition

Deep Learning is a subset of Machine Learning that uses Artificial Neural Networks (ANNs) with multiple layers to learn complex patterns from data. It is inspired by the structure and functioning of the human brain.

Unlike traditional machine learning, deep learning automatically extracts features from raw data without manual intervention.

Example

In face recognition:

- Traditional ML requires manual extraction of features such as eyes, nose, and mouth.
- Deep Learning automatically learns these features from images.

Advantages

- High accuracy
- Automatic feature extraction
- Handles large and complex datasets

Disadvantages

- Requires large amounts of data
- High computational cost
- Longer training time

Applications

- Face Recognition
 - Speech Recognition
 - Self-Driving Cars
 - Medical Diagnosis
-

2. Supervised Learning

Definition

Supervised Learning is a learning method where the training dataset contains both input data and corresponding output labels.

The model learns a mapping between inputs and outputs.

Example

Input Image Label

Dog Image Dog

Cat Image Cat

The model learns from labeled examples and predicts labels for new images.

Process

1. Collect labeled data.
2. Train the model.
3. Validate performance.
4. Make predictions.

Applications

- Image Classification
- Spam Detection
- Medical Diagnosis
- Face Recognition

Advantages

- High prediction accuracy
- Easy performance evaluation

Disadvantages

- Requires large labeled datasets
 - Data labeling is expensive and time-consuming
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3. Unsupervised Learning

Definition

Unsupervised Learning is a learning method where the dataset contains no labels.

The model discovers hidden structures and patterns in data.

Example

Given thousands of images, the model automatically groups similar images together without knowing their categories.

Applications

- Customer Segmentation
- Data Clustering
- Pattern Discovery
- Market Analysis

Common Techniques

- Clustering
- Dimensionality Reduction

Advantages

- No labeled data required
- Useful for discovering unknown patterns

Disadvantages

- Results may be difficult to interpret
 - Lower accuracy compared to supervised learning
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4. Deep Neural Networks (DNN)

Definition

A Deep Neural Network is an Artificial Neural Network containing multiple hidden layers between the input and output layers.

Architecture

Input Layer

↓

Hidden Layer 1

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Hidden Layer 2

↓

Hidden Layer 3

↓

Output Layer

Working

Each hidden layer extracts increasingly complex features.

For image recognition:

- Layer 1 detects edges.
- Layer 2 detects shapes.
- Layer 3 detects object parts.
- Output layer identifies the object.

Advantages

- Learns highly complex patterns.

- Suitable for large datasets.

Applications

- Image Recognition
 - Speech Processing
 - Natural Language Processing
-

5. Convolutional Neural Networks (CNN)

Definition

CNN is a specialized Deep Neural Network designed for image and video processing.

It automatically extracts important features from images.

CNN Architecture

1. Convolution Layer

Extracts features such as:

- Edges
- Corners
- Textures
- Shapes

Small filters (kernels) slide across the image.

2. ReLU Layer

Applies an activation function.

Purpose:

- Removes negative values.
- Introduces non-linearity.

3. Pooling Layer

Reduces image dimensions.

Types:

- Max Pooling
- Average Pooling

Benefits:

- Reduces computation
- Prevents overfitting

4. Fully Connected Layer

Performs final classification based on extracted features.

Workflow

Image

↓

Convolution

↓

ReLU

↓

Pooling

↓

Fully Connected Layer

↓

Output Class

Applications

- Face Detection
- Medical Imaging
- Self-Driving Cars
- Object Recognition

Advantages

- Automatic feature extraction
 - High classification accuracy
-

6. More Complex Models

Advanced deep learning architectures include:

ResNet (Residual Network)

Uses skip connections to solve the vanishing gradient problem.

VGGNet

Uses many convolution layers for improved accuracy.

Inception Network

Performs multiple convolution operations simultaneously.

Vision Transformers (ViT)

Uses transformer architecture for image analysis.

Recognition

Recognition refers to identifying and understanding objects, people, or scenes within images and videos.

7. Instance Recognition

Definition

Instance Recognition identifies a specific object instance rather than only its category.

Example

Category Recognition:

- "Dog"

Instance Recognition:

- "Tommy the Dog"

Applications

- Face Recognition
 - Product Recognition
 - Landmark Recognition
-

8. Image Classification

Definition

Image Classification assigns a single label to an entire image.

Example

Input:

Image of a cat

Output:

Cat

Types

Binary Classification

Two classes.

Example:

- Cancer
- No Cancer

Multi-Class Classification

More than two classes.

Example:

- Dog
- Cat
- Horse
- Bird

Applications

- Medical Diagnosis
 - Plant Identification
 - Animal Recognition
-

9. Object Detection

Definition

Object Detection identifies objects and their locations within an image.

Output:

- Object Label
- Bounding Box

Example

An image contains:

- Car
- Person
- Dog

Each object receives a separate bounding box.

Popular Algorithms

- YOLO
- R-CNN
- Faster R-CNN
- SSD

Applications

- Surveillance Systems
 - Autonomous Vehicles
 - Security Monitoring
-

10. Semantic Segmentation

Definition

Semantic Segmentation classifies every pixel in an image.

Example

Road Scene:

Pixels are labeled as:

- Road
- Car
- Building
- Tree

Difference

Method	Output
Classification	Whole image label
Detection	Object location
Segmentation	Pixel-level classification

Applications

- Autonomous Driving
- Medical Imaging
- Satellite Image Analysis

11. Pose Estimation

Definition

Pose Estimation detects the positions of body joints.

Key Points Detected

- Head
- Shoulders
- Elbows
- Knees
- Ankles

Applications

- Sports Analytics
- Fitness Tracking
- Human-Computer Interaction
- Motion Capture

12. Video Understanding

Definition

Video Understanding analyzes video content and recognizes activities or events.

Tasks

- Action Recognition
- Event Detection
- Activity Classification

Example

Video:

Person running

Output:

Running

Applications

- CCTV Monitoring
- Sports Analysis
- Smart Surveillance

13. Vision and Language

Definition

Combines Computer Vision and Natural Language Processing (NLP).

Example

Input:

Image of a dog playing in a park

Output:

"A dog is playing in a park."

Applications

- Image Captioning
- Visual Question Answering
- AI Assistants

Feature Detection and Matching

Feature detection identifies important structures in images.

14. Edges and Contours

Edge

An edge is a sudden change in image intensity.

Edge Detection Methods

- Sobel Operator
- Canny Edge Detector
- Laplacian Operator

Applications

- Object Detection
 - Shape Analysis
-

Contour

A contour is the boundary of an object.

Applications

- Shape Recognition
 - Object Tracking
 - Image Analysis
-

15. Contour Tracking

Definition

Tracking object boundaries across multiple video frames.

Applications

- Video Surveillance
 - Motion Tracking
 - Traffic Monitoring
-

16. Lines and Vanishing Points

Line Detection

Lines are detected using techniques such as:

Hough Transform

Used for:

- Road lane detection
- Building structure analysis

Vanishing Point

A point where parallel lines appear to meet due to perspective.

Example

Railway tracks appear to converge at a distant point.

Applications

- 3D Reconstruction
 - Autonomous Driving
 - Camera Calibration
-

17. Segmentation

Definition

Segmentation divides an image into meaningful regions.

Types

Threshold-Based Segmentation

Uses intensity threshold values.

Region-Based Segmentation

Groups neighboring pixels with similar properties.

Edge-Based Segmentation

Uses detected edges to define regions.

Clustering-Based Segmentation

Uses algorithms such as K-Means.

Applications

- Medical Imaging
 - Object Detection
 - Satellite Analysis
-

Motion Estimation

Motion estimation determines movement between video frames.

18. Translational Alignment

Definition

Aligning images using horizontal or vertical shifts.

Applications

- Panorama Generation
 - Image Registration
 - Image Stitching
-

19. Parametric Motion

Definition

Represents motion using mathematical parameters.

Parameters

- Translation
- Rotation
- Scaling
- Shearing

Applications

- Video Stabilization
 - Motion Analysis
-

20. Optical Flow

Definition

Optical Flow estimates the motion of pixels between consecutive frames.

Example

Frame 1:

Ball at left position

Frame 2:

Ball at right position

Optical Flow calculates motion vectors showing movement.

Applications

- Object Tracking
- Gesture Recognition
- Autonomous Vehicles
- Video Compression

Advantages

- Accurate motion estimation

- Useful for dynamic scene analysis
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21. Layered Motion

Definition

Represents motion by separating a scene into multiple moving layers.

Example

Foreground:
Moving person

Background:
Static building

Applications

- Video Editing
 - Motion Analysis
 - Scene Understanding
-

Computational Photography

Computational Photography uses algorithms to improve image quality beyond traditional camera capabilities.

22. High Dynamic Range (HDR) Imaging

Definition

HDR combines multiple images captured at different exposure levels.

Purpose

Preserves details in:

- Bright regions
- Dark regions

Applications

- Smartphone Cameras
- Professional Photography

Advantages

- Improved contrast
 - Better image quality
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23. Super-Resolution

Definition

Converts low-resolution images into high-resolution images.

Example

480p → 1080p

Applications

- CCTV Enhancement
 - Medical Imaging
 - Satellite Imaging
-

24. Denoising**Definition**

Removing unwanted noise from images.

Common Filters

- Mean Filter
- Median Filter
- Gaussian Filter

Applications

- Medical Images
 - Photography
 - Satellite Images
-

25. Blur Removal (Deblurring)**Definition**

Restoring sharpness in blurred images.

Causes of Blur

- Camera shake
- Object motion
- Poor focus

Applications

- Security Cameras
 - Mobile Photography
 - Scientific Imaging
-

26. Image Matting and Compositing

Image Matting

Separates foreground objects from the background.

Example

Extracting a person from an image.

Image Compositing

Combines extracted foreground with a new background.

Applications

- Photoshop
 - Movie Special Effects
 - Advertising
-

27. Video Matting

Definition

Separating foreground objects from the background in every video frame.

Applications

- Green Screen Effects
 - Virtual Backgrounds
 - Video Editing
-

28. Texture Analysis

Definition

Analyzing surface patterns in images.

Examples

- Wood texture
- Fabric texture
- Grass texture

Applications

- Medical Diagnosis
 - Material Identification
 - Industrial Inspection
-

29. Texture Synthesis

Definition

Generating larger or new textures from sample textures.

Example

Creating a large brick wall texture from a small sample.

Applications

- Computer Graphics
 - Gaming
 - Animation
 - Virtual Reality
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